

Liver Disease Detection Using Deep Learning (ANN)

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Abstract

The detection of liver disease at an early stage is critical for ensuring timely medical intervention. This project focuses on developing an automated liver disease detection system using a deep learning approach, utilizing both Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN). The dataset consists of 2,391 patient medical records, combining Kaggle data, lab reports (from Maharashtra), and records generated through AI techniques. The dataset includes features such as age, gender, bilirubin levels, enzyme concentrations, and protein ratios, all indicative of liver function. Extensive data preprocessing was conducted, including handling missing values, encoding categorical variables, removing outliers, and feature scaling. The ANN model, comprising multiple hidden layers with ReLU activation, was trained to classify patients as either having liver disease or not. The model's performance was evaluated using metrics such as accuracy, precision, recall, and F1-score, achieving an accuracy of 92% on the test dataset. Further improvements are possible through advanced architectures, hyperparameter tuning, and data augmentation. This research underscores the viability of deep learning in facilitating early and non-invasive liver disease detection.

Keywords: Liver disease, Deep learning, ANN, CNN, Healthcare, Data Science.

I. Introduction

This chapter provides an overview of liver disease, its prevalence, and the challenges of early detection. It introduces the motivation behind using Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) for liver disease diagnosis and outlines the project's aim to develop a non-invasive, data-driven detection model. The chapter also presents the problem statement, project scope, and specific objectives for classification accuracy. By analyzing various patient indicators, this project demonstrates how ANN and CNN models effectively classify individuals based on liver health [1][2].

The liver, located in the upper gastrointestinal tract, weighs between 1,200–1,800 grams and is vital for digestion, metabolism, detoxification, immune function, and nutrient storage [1]. Liver diseases, which can lead to serious health issues and even death, are categorized based on their causes—such as infections, injuries, genetic abnormalities, or exposure to toxins [3]—and their effects on liver function [4]. Key liver diseases include non-alcoholic fatty liver disease (NAFLD) [6], cirrhosis [5], and various forms of hepatitis [12]. NAFLD is characterized by lipid accumulation, often linked to obesity and diabetes [20]. Cirrhosis involves the replacement of healthy liver tissue with scar tissue, frequently caused by alcoholism or chronic hepatitis [17]. Hepatitis can be acute or chronic and is primarily caused by viral infections from a group of hepatitis viruses (A, B, C, D, and E) [13], each with distinct transmission methods [15]. Common liver diseases such as NAFLD, cirrhosis, and hepatitis have different health implications [3][4]. Advanced stages of fatty liver disease can significantly elevate mortality risk [19]. Preventative measures include reducing alcohol consumption [18], practicing safe sex [14], avoiding shared needles [16], getting vaccinated against hepatitis A and B [11], and maintaining a healthy lifestyle [10]. Traditional methods for diagnosing liver disease are often invasive and costly, emphasizing the need for non-invasive solutions like deep learning models [6][22]. Increasingly, liver disease is linked to obesity and diabetes [21], posing a silent but severe health threat. With advancements in artificial intelligence (AI) and machine learning (ML) [23], healthcare professionals are improving disease detection through better data collection and analysis [24]. Emerging research suggests that gut microbiota may play a significant role in liver disease progression and management [26]. Dysbiosis, or an imbalance in gut bacteria, has been linked to NAFLD and cirrhosis, highlighting potential therapeutic interventions through probiotics and dietary modifications [27]. Additionally, genetic predisposition is increasingly recognized as a contributing factor to liver diseases, with genome-wide association studies (GWAS) identifying specific genes linked to liver dysfunction [28].

This chapter provides an overview of liver disease, its prevalence, and the challenges of early detection. It introduces the motivation behind using Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) for liver disease diagnosis and outlines the project's aim to develop a non-invasive, data-driven detection model. The chapter also presents the problem statement, project scope, and specific objectives for classification accuracy. This report highlights the potential of ANN and CNN for liver disease detection, focusing on classifying patients based on crucial features such as age [39], bilirubin levels [40], liver enzyme levels [42], and other liver indicators [45]. By integrating these technologies, clinicians can enhance diagnostic accuracy and improve patient outcomes in liver health management [25].

II. Dataset Description

The dataset utilized in this study comprises 2,391 patient records, derived from three sources:

1. A publicly available dataset from Kaggle[7].
 2. Collected from local Laboratories from Maharashtra.
 3. Synthetic data generated using AI techniques to ensure diversity and balance in class distribution.
- Each record contains the following features:
 - **Demographics:** Age, Gender
 - **Liver Function Indicators:**
 - Bilirubin levels (Total and Direct)
 - Enzyme concentrations (e.g., Alanine Transaminase (ALT), Aspartate Transaminase (AST))
 - Protein levels (e.g., Albumin, Globulin ratio)

Table:1 Dataset Features

Feature	Type	Description
Gender [38]	Nominal	This feature illustrates the participant's gender.
Age (years) [39]	Numeric	The age range of the participants is 4–90 years.
Total Bilirubin—TB (mg/dL) [40]	Numeric	This feature captures the participant's
Direct Bilirubin—DB (mg/dL) [40]	Numeric	This feature captures the participant's direct bilirubin.
Alkaline Phosphatase—ALP (IU/L) [41]	Numeric	This feature captures the participant's alkaline phosphatase.
Alanine Aminotransferase—SGPT (U/L) [42]	Numeric	This feature captures the participant's alanine aminotransferase.
Aspartate Aminotransferase—SGOT (U/L) [42]	Numeric	This feature captures the participant's aspartate aminotransferase.
Total Protein—TP (g/L) [43]	Numeric	This feature captures the participant's total protein.
Albumin—ALB (g/dL) [44]	Numeric	This feature captures the participant's albumin.
Albumin and Globulin Ratio—AGR [45]	Numeric	This feature captures the participant's albumin and globulin Ratio.
Liver Disease	Nominal	This feature stands for whether the participant has been diagnosed with liver disease or not.

III. Methodology

In this study, we proposed a deep learning-based approach for data classification. The methodology follows a structured pipeline that includes data preprocessing, feature selection, deep learning model training, classification, and result evaluation.

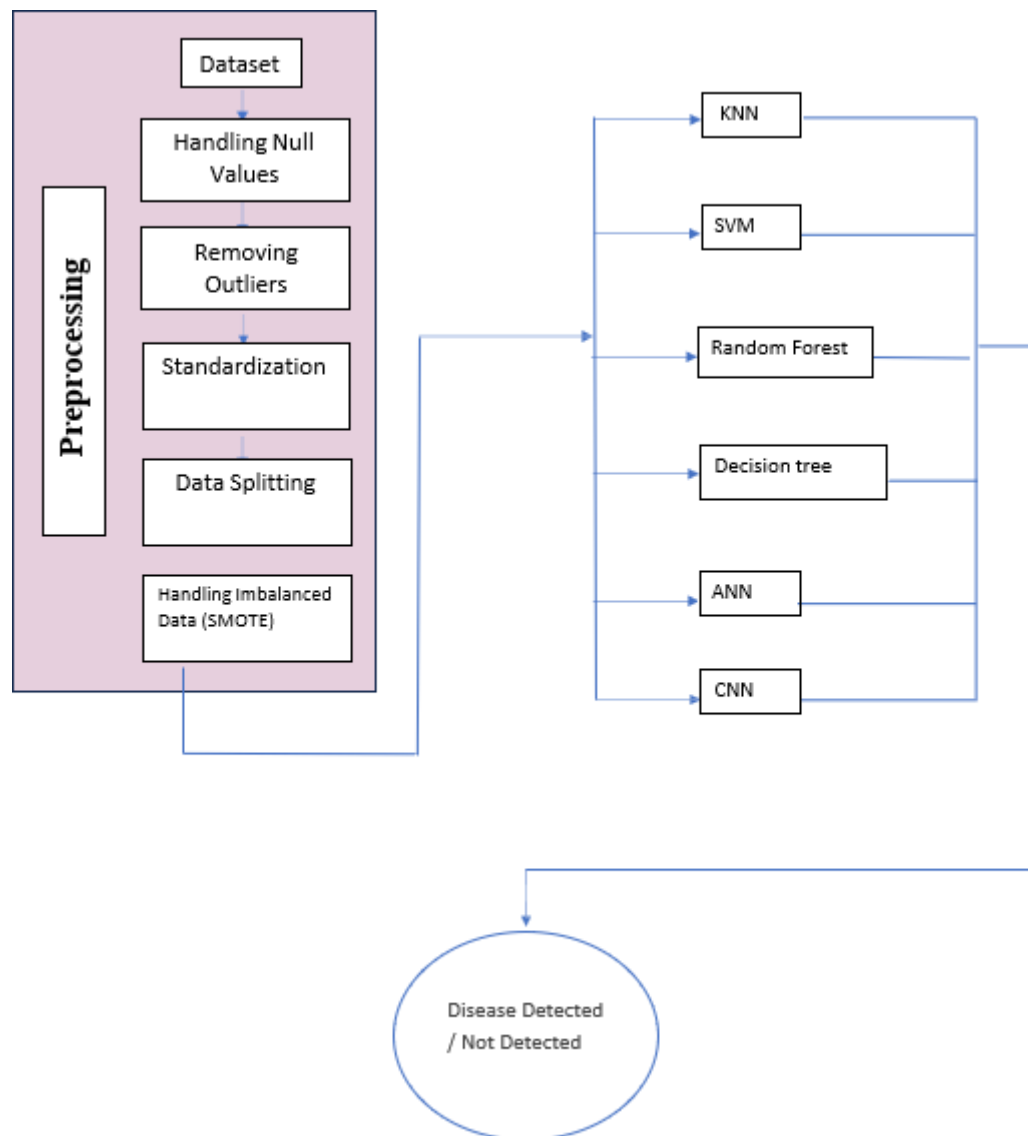


Fig.1 Methodology

- **Data Collection and Preprocessing:** The dataset used for this study consists of 2391 patient records. Each record includes demographic and clinical features relevant to liver function, such as age, gender, bilirubin levels, enzyme concentrations, and protein ratios. To ensure data quality, the preprocessing steps involved:

- **Handling Missing Values:**

One of the first steps in data preprocessing is to deal with missing or incomplete data. Missing data can occur due to various reasons, such as errors in data collection or patients not undergoing specific tests. Handling missing values is crucial because most machine learning models, including ANNs, cannot handle missing data directly. In this dataset, we conducted an extensive check for missing values using pandas functions. Fortunately, the dataset had no missing values in any of the features. If missing data had been present, several imputation strategies could have been applied:

- **Mean/Median Imputation:** For continuous variables, replacing missing values with the mean or median of the available data.
- **Mode Imputation:** For categorical variables like gender, missing values could be replaced with the most frequent category.

- **Removing Outliers:**

To identify outliers, we used a Box Plot to visualize the distribution of the data and detect extreme values. We then applied the Interquartile Range (IQR) method to systematically remove outliers. The IQR method involves calculating the range between the first quartile (Q1) and third quartile (Q3), and using this range to identify and eliminate data points that lie outside the acceptable limits, typically beyond 1.5 times the IQR. After outliers were removed, imputation was performed on missing values to ensure that the dataset was complete and usable for further analysis. Depending on the nature of the data, different imputation methods, such as mean, median, or regression-based imputation, were applied to ensure consistency and preserve data integrity.

These preprocessing steps are essential to ensure that the data is clean, reliable, and ready for the subsequent analysis and modelling stages. By addressing outliers and missing data, we improve the quality of the data, reduce noise, and enhance the accuracy of any machine learning models or statistical analysis performed.

- **Feature Scaling:**

Artificial Neural Networks (ANNs) are sensitive to the scale of input features. If one feature has a much larger range than others, it can dominate the learning process and lead to suboptimal performance. For example, the Age feature ranges from 0 to 90, whereas Albumin levels are in a much smaller range, around 1-5. To ensure all features contribute equally, feature scaling was applied.

- **Standardization:** In this project, standardization (z-score normalization) was applied, which transforms each feature to have a mean of 0 and a standard deviation of 1. This is done using the formula:

- $$z = \frac{(X - \mu)}{\sigma}$$

Where,

- x is the original value,
- μ is the mean of the feature,
- σ is the standard deviation of the feature.

Standardization ensures that all the features have the same scale, regardless of their original ranges. This is particularly important for distance-based algorithms and neural networks, where feature magnitudes can affect the learning process. The **Standard Scaler** from the scikit-learn library was used to scale the dataset. The scaler was fit on the training data and then applied consistently to both the training and test data to ensure that the test data was processed in the same way.

- **Data Splitting:**

To evaluate the performance of both the **Artificial Neural Network (ANN)** and **Convolutional Neural Network (CNN)** models, the dataset was split into **training** and **test** sets. This is a standard practice in machine learning to ensure that the model can generalize well to unseen data.

- **Train-Test Split:** We used the **train_test_split** function from the **scikit-learn** library to split the dataset into 80% training data and 20% test data. This 80/20 ratio is commonly adopted in machine learning projects as it provides enough data for training while keeping a sufficient portion for testing the model's generalization ability.

The **training set** is used to teach the model, whether it's the ANN or CNN, by learning the patterns and relationships in the data. The **test set** is reserved to evaluate how well the trained model performs on **unseen data**, ensuring that it can generalize to new, real-world cases.

The use of **CNN** is particularly useful for tasks like image classification, object detection, or any task where spatial hierarchies and patterns are important. CNNs apply convolutional filters to detect local features, such as edges, textures, and shapes, making them highly effective in visual data analysis.

By splitting the data into training and test sets, we can prevent the models from overfitting, where they might memorize the data instead of learning the underlying patterns. The test set provides a realistic evaluation of how the model would perform in real-world scenarios when deployed.

Additionally, using both ANN and CNN allows us to compare the performance of traditional neural networks (ANNs) and more specialized deep learning models (CNNs) in handling the dataset, helping us to choose the most effective model for the task at hand.

• **Handling Imbalanced Data (SMOTE)**

In many machine learning tasks, especially classification problems, the dataset may have an **imbalanced class distribution**, where one class has significantly more instances than the other. This imbalance can lead to biased model performance, as the model may become biased toward predicting the majority class and underperform in predicting the minority class.

To address this issue, we used the **Synthetic Minority Over-sampling Technique (SMOTE)**. SMOTE is an effective technique for handling imbalanced data by generating synthetic examples for the minority class. Unlike simple oversampling, which involves duplicating existing minority class examples, SMOTE creates new, plausible instances by interpolating between existing data points in the feature space.

• **How SMOTE works:**

- SMOTE selects instances from the minority class.
- For each selected instance, it finds its nearest neighbors.
- A synthetic instance is created by taking a random point along the line segment between the selected instance and one of its neighbors.

- This process is repeated until the minority class is sufficiently balanced with the majority class.

This approach helps in several ways:

1. **Improved Model Performance:** By balancing the dataset, SMOTE prevents the model from being biased toward the majority class and helps improve its ability to detect patterns related to the minority class.
2. **Better Generalization:** The synthetic samples created by SMOTE help the model generalize better to unseen data, improving its performance in real-world scenarios where imbalanced class distributions are common.
3. **Reduced Overfitting:** Unlike simple oversampling, SMOTE reduces the risk of overfitting since it creates new instances instead of just replicating the existing data. We applied SMOTE on the training set after the initial train-test split to ensure that the model learns a balanced representation of both classes. This step was essential in cases where the model's predictions for the minority class were initially poor, ensuring fairer and more accurate classification results.

• Model Architecture:

The Artificial Neural Network (ANN) model was designed to capture complex patterns in the dataset. The network structure consists of:

- Input Layer: Accepts 10 features representing patient demographics and clinical biomarkers.
- Hidden Layers:
 - First hidden layer: 64 neurons with Rectified Linear Unit (ReLU) activation.
 - Second hidden layer: 32 neurons with ReLU activation.
- Output Layer: A single neuron with a sigmoid activation function to classify patients into two categories: with or without liver disease.

Table. 02 Artificial Neural Network Model Architecture

Layer Type	Layer Details
Dense (Layer 1)	Units: 10, Input Shape: (num_features,), Activation: ReLU
Dense (Layer 2)	Units: 20, Activation: ReLU
Dense (Layer 3)	Units: 40, Activation: ReLU
Dense (Layer 4)	Units: 80, Activation: ReLU
Dense (Layer 5)	Units: 1, Activation: Sigmoid

Table. 03 Artificial Neural Network Model Architecture with L1-L2 Regularization

Layer Type	Layer Details
Dense (Layer 1)	Units: 10, Activation: ReLU, Input Shape: (num_features,), Regularization: L1=0.01, L2=0.01
Dense (Layer 2)	Units: 20, Activation: ReLU, Regularization: L1=0.01, L2=0.01
Dense (Layer 3)	Units: 40, Activation: ReLU, Regularization: L1=0.01, L2=0.01
Dense (Layer 4)	Units: 80, Activation: ReLU, Regularization: L1=0.01, L2=0.01
Dense (Output Layer)	Units: 1, Activation: Sigmoid

Table. 05 Artificial Neural Network Model Architecture with Dropout

Layer Type	Layer Details
Dense (Layer 1)	Units: 10, Activation: ReLU, Input Shape: (num_features,)
Dropout (Layer 1)	Rate: 0.2
Dense (Layer 2)	Units: 20, Activation: ReLU
Dropout (Layer 2)	Rate: 0.2
Dense (Layer 3)	Units: 40, Activation: ReLU
Dropout (Layer 3)	Rate: 0.2
Dense (Layer 4)	Units: 80, Activation: ReLU
Dropout (Layer 4)	Rate: 0.2
Dense (Output Layer)	Units: 1, Activation: Sigmoid

Table. 0 Convolutional Neural Network Model Architecture

Layer Type	Layer Details
Conv1D (Layer 1)	Filters: 32, Kernel Size: 3, Activation: ReLU, Input Shape: (X_train_cnn.shape[1], 1)
MaxPooling1D (Layer 1)	Pool Size: 2
Conv1D (Layer 2)	Filters: 64, Kernel Size: 2, Activation: ReLU, Padding: 'same'
MaxPooling1D (Layer 2)	Pool Size: 2
Flatten	-
Dense	Units: 1, Activation: Sigmoid

- Model Training and Optimization:**

The ANN model was trained using the following configurations:

- Loss Function: Binary Cross-Entropy, suitable for binary classification tasks.

- **Optimizer:** Adam optimizer, chosen for its adaptive learning rate capabilities.
- **Batch Size:** 32, to balance training efficiency and stability.
- **Epochs:** 100, determined based on the model's convergence behavior.
- **Regularization Techniques:** Dropout layers were introduced to mitigate overfitting by randomly deactivating neurons during training.

- **Model Evaluation:**

The trained model was assessed using various performance metrics to ensure its effectiveness:

- **Accuracy:** The proportion of correctly classified instances.
- **Precision:** The ratio of true positive predictions to the total predicted positives.
- **Recall:** The ability of the model to identify actual cases of liver disease.
- **F1-score:** The harmonic mean of precision and recall, balancing both metrics.
- **Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC):** Evaluated the model's ability to differentiate between positive and negative cases. The model achieved an AUC score of 0.92, indicating high classification ability.

- **Testing and Validation:**

The trained model was tested on the unseen 20% test dataset to assess its generalization capability. The validation loss and accuracy trends were monitored across epochs to detect potential overfitting.

IV.Result:

Evaluation of Deep Learning Models

Model	Training Accuracy	Testing Accuracy	Training Precision	Testing Precision	ROC Area
ANN	93%	91%	95%	93%	0.97
ANN (L2 Regularization)	91%	91%	95%	93%	0.97
ANN (L1L2 Regularization)	91%	91%	93%	93%	0.97
ANN (Dropout 20%)	92%	92%	93%	92%	0.97
CNN (2 Conv Layers)	92%	91%	95%	92%	0.97
CNN (3 Conv Layers)	92%	91%	96%	93%	0.98

- Training Accuracy indicates how well the model fits the training data. Most models achieved high training accuracy, with the ANN models generally performing slightly better in this regard.
- Testing Accuracy shows the model's ability to generalize to new, unseen data. Both ANN and CNN models achieved similar testing accuracy, with most models performing around 91% to 92% on the test set. This suggests that the models are capable of generalizing well.
- Training Precision and Testing Precision provide insight into how well the models perform when predicting the positive class (in a classification context). Precision values are quite high across all models, ranging from 93% to 96%, suggesting that the models are good at making correct positive class predictions.
- ROC Area is an important metric for evaluating the model's ability to discriminate between classes. A value closer to 1 indicates better performance. Most models had an ROC area around 0.97, with the CNN model with 3 Conv Layers slightly outperforming the others with an ROC area of 0.98.

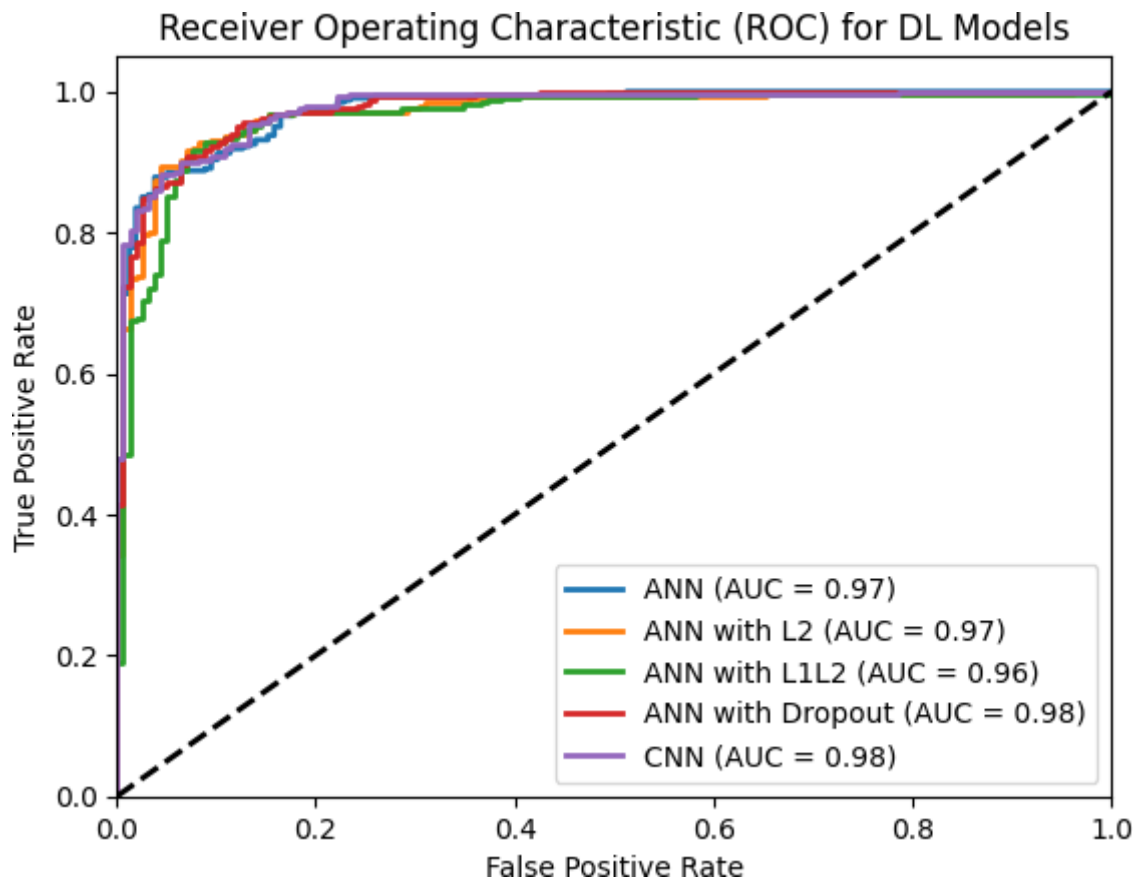


Fig.2 Receiver Operating Characteristic (ROC) for DL Models

This graph represents the Receiver Operating Characteristic (ROC) curves for various Deep Learning (DL) models used for classification. The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) at different classification thresholds.

Key Elements of the Graph

1. X-Axis (False Positive Rate - FPR):
 - Measures the proportion of incorrectly classified negative samples as positive.
 - Lower values indicate better model performance.
2. Y-Axis (True Positive Rate - TPR):
 - Measures the proportion of correctly classified positive samples.
 - Higher values indicate better model performance.
3. Diagonal Line (Black Dashed Line):
 - Represents a random classifier with no discrimination capability (AUC = 0.5).
 - Any model above this line is performing better than random chance.
4. Colored ROC Curves:
 - Each curve represents a different DL model's performance.
 - Models closer to the top-left corner indicate better performance.

Models and Their AUC Scores

- ANN (Artificial Neural Network) (AUC = 0.97): A standard ANN model.
- ANN with L2 Regularization (AUC = 0.97): Regularization helps prevent overfitting.
- ANN with L1L2 Regularization (AUC = 0.96): Uses both L1 and L2 norms to penalize weights.
- ANN with Dropout (AUC = 0.98): Dropout is used to prevent overfitting by randomly deactivating neurons.
- CNN (Convolutional Neural Network) (AUC = 0.98): A CNN model that slightly outperforms the standard ANN.

Observations

- The CNN and ANN with Dropout achieved the highest AUC (0.98), indicating better classification performance.
- The standard ANN and ANN with L2 regularization performed slightly lower (AUC = 0.97).
- The ANN with L1L2 regularization has the lowest AUC (0.96), but still performs well.

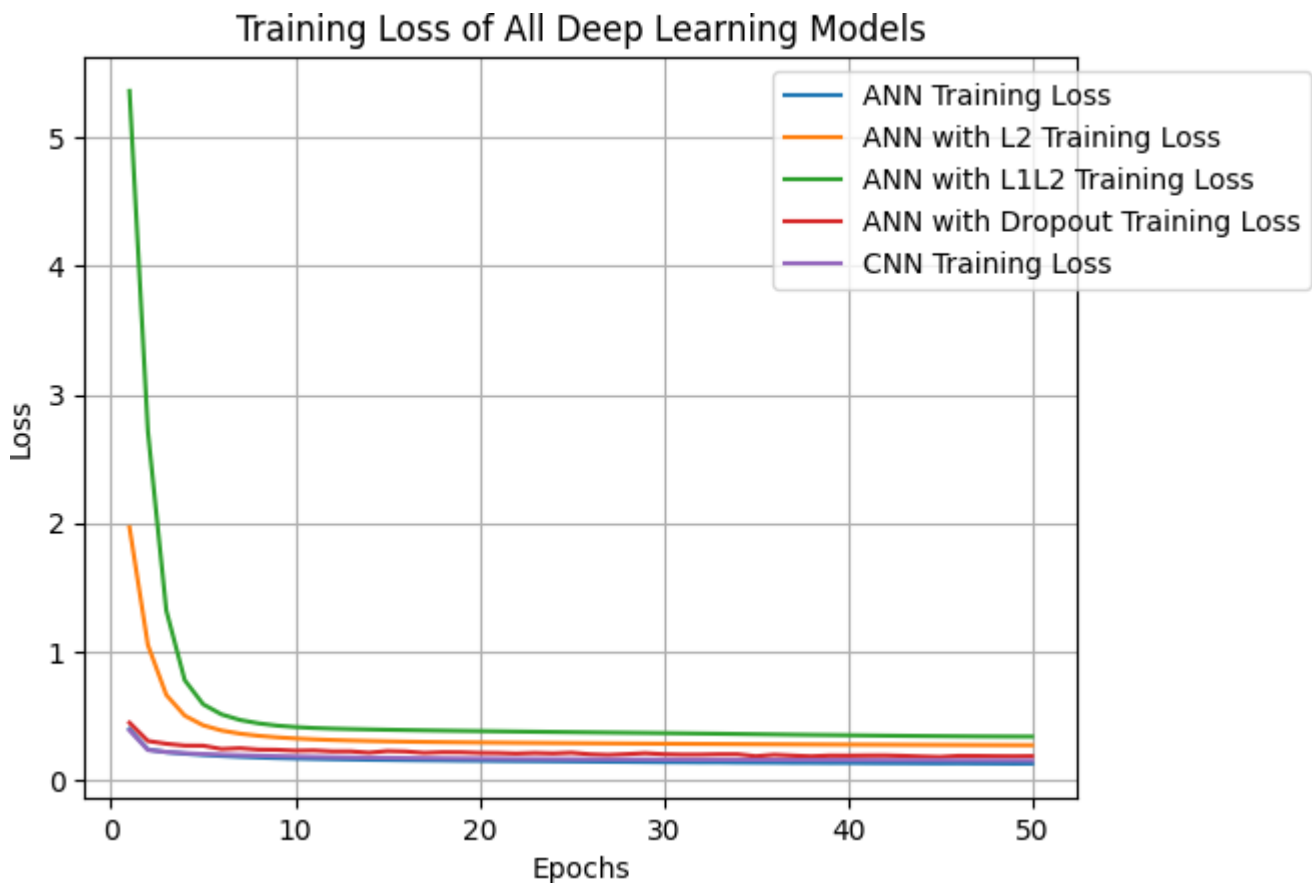


Fig.3 Training Loss of All Deep Learning Models

This graph represents the training loss curves of different Deep Learning (DL) models over 50 epochs. Training loss measures how well a model is learning by evaluating the difference between predicted and actual values.

Key Elements of the Graph

1. X-Axis (Epochs):

- Represents the number of training iterations.
- As epochs increase, the model adjusts weights to reduce the error.

2. Y-Axis (Loss):

- Represents the loss value (e.g., categorical cross-entropy or MSE).
- A lower loss indicates better model performance.

3. Colored Loss Curves:

- Each curve represents the training loss of a different DL model.
- A steep drop at the beginning is common, as models learn patterns quickly in early epochs.

Models and Their Loss Curves

- ANN Training Loss (Blue Curve): A standard ANN that converges smoothly.
- ANN with L2 Regularization (Orange Curve): Has a slightly higher initial loss but stabilizes.
- ANN with L1L2 Regularization (Green Curve): Starts with the highest loss (~5.5) but eventually converges.
- ANN with Dropout (Red Curve): Shows a stable loss curve.
- CNN Training Loss (Purple Curve): Has the lowest loss among all models, indicating better learning.

Observations

- All models show a decreasing trend, which means they are learning properly.
- The ANN with L1L2 regularization started with the highest loss, but it eventually converges, showing that the model required more updates to optimize.
- CNN has the lowest loss, indicating that it learns patterns more effectively.
- Regularized models (L2, L1L2, Dropout) tend to have slightly higher loss values due to the penalty applied to weights, which helps prevent overfitting.

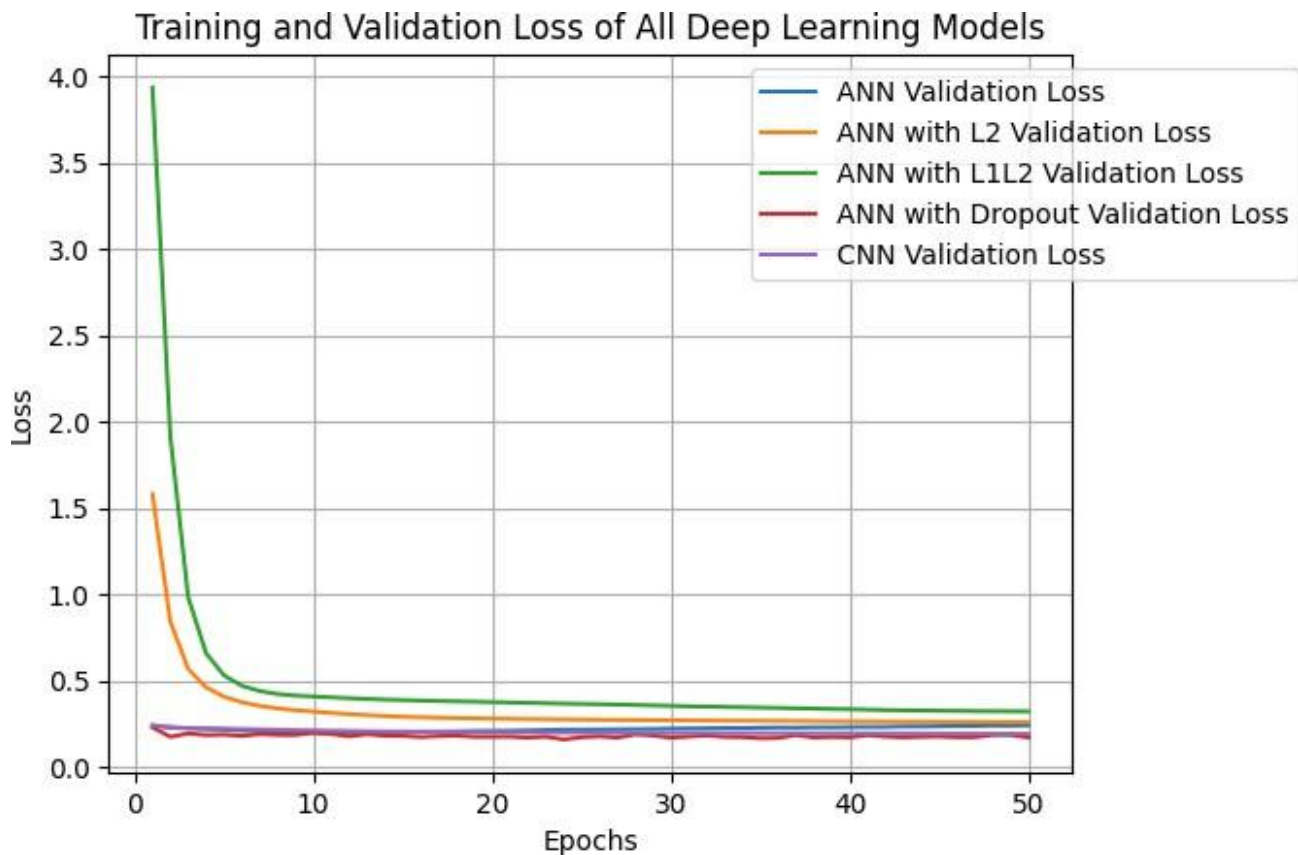


Fig.4 Testing Loss of All Deep Learning Models

This graph represents the training and validation loss of multiple deep learning models over 50 epochs. Here's what the graph indicates:

Axes:

- X-axis (Epochs): Represents the number of training iterations (1 to 50).
- Y-axis (Loss): Measures the error in predictions, with lower values indicating better model performance.

Curves in the Graph:

- Blue Line: Represents the validation loss of a standard Artificial Neural Network (ANN).
- Orange Line: Represents the validation loss of an ANN with L2 regularization (helps reduce overfitting by adding a penalty to large weights).
- Green Line: Represents the validation loss of an ANN with L1L2 regularization (combination of L1 and L2 penalties).
- Red Line: Represents the validation loss of an ANN with Dropout regularization (randomly drops neurons to prevent overfitting).
- Purple Line: Represents the validation loss of a Convolutional Neural Network (CNN).

Observations:

1. Initial High Loss:
 - All models start with a high loss, but it quickly decreases within the first 10 epochs.
2. Stabilization:
 - After around 10–20 epochs, the loss values flatten, indicating that models are converging.
3. Performance Comparison:
 - The green curve (ANN with L1L2) starts with the highest loss but eventually stabilizes.
 - The red and purple curves (Dropout ANN and CNN) show the lowest loss, indicating that these models generalize better.
 - The CNN (purple) seems to perform better than all ANN models in terms of maintaining a low loss.

○

V. Discussion

The proposed model highlights the feasibility of using deep learning for liver disease detection. Key strengths include:

- Effective handling of class imbalance through synthetic data generation.
- Robust performance metrics, with high precision and recall.

VI. Conclusion

This research demonstrates the potential of Artificial Neural Networks in liver disease detection, providing a non-invasive, efficient alternative to traditional methods. With an accuracy of 92 %, the model shows promise for early-stage diagnosis, paving the way for future advancements in healthcare technology. Further research into IoT integration, larger datasets, and advanced architectures could significantly enhance its applicability.

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